Statistical protocol: matched comparisons, intervals, and bounded claims

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Statistical protocol (part 2)

No “robust beats naive” claim is written before a statistical test produces it (Algorithm Gate I). Among the active-inference sources reviewed here, belief-fusion outcomes are typically reported as single illustrative runs; we treat every headline contrast as a paired hypothesis test with effect-size estimation,

Statistical protocol (part 3)

multiple-testing deflation, confidence intervals,
and an observed-effect design-power calculation —

Statistical protocol (part 4)

the reporting discipline expected by robust statistics and applied inference (Huber and Ronchetti 2009; Efron and Tibshirani 1993; Koehler et al. 2009; Morris et al. 2019; Loy and Korobova 2021; Nakagawa and Cuthill 2007; Benjamini and Hochberg 1995; Wasserstein and Lazar 2016).

Statistical protocol (part 5)

The protocol lives in `statistics.py`, sits at the analysis tier above the locked FedGVI core, and emits every number the results sections report; nothing is hard-coded (ISC-30).

Paired comparison and standardized effect size

The headline claim is *paired*: across matched scenarios — same seed, same contamination — does the robust aggregator raise consensus accuracy over the naive log-linear pool? For the expanded review grid,

Paired comparison and... (part 2)

the inferential unit is the seed: trials are nested within a seed/cell and are averaged before seed-level contrasts.

Paired comparison and... (part 3)

The primary robustness sweep also retains a trial-level paired diagnostic, but its matched trial replicates are nested within one fixed seeded world and are not independent worlds. The seed-level review-grid estimand is therefore the robust-minus-naive accuracy difference after the declared trial reduction,

Paired comparison and... (part 4)

not a contrast of two independently drawn group means.

Paired comparison and... (part 5)

Each configured robust method is retained in every review-grid rate panel and has its own seed-level contrast, interval, and test; no pooled-selected curve or pooled-selected inferential member enters that grid.

Paired comparison and... (part 6)

The naive comparator is the project's log-linear pool eq. 9. Under the shared-support, posterior-log-potential, and fixed-weight assumptions of sec. 8.8, it specializes the message-combination term of Friston et al.'s Eq. 7 (2024),

Paired comparison and... (part 7)

rather than reconstructing the complete source protocol.

Paired comparison and... (part 8)

We test it with the matched-pairs Wilcoxon signed-rank test (Wilcoxon 1945) (`statistics.paired_test`, ISC-28), interpreted under the usual signed-rank conditions: seed-level paired differences are independent across the declared seed schedule,

Paired comparison and… (part 9)

while the primary sweep's trial-level diagnostic remains conditional on its fixed world; the differences are ranked after zero differences are removed, and the null is a symmetric distribution of paired differences around zero rather than an equality-of-means claim (Fay and Proschan 2010).

Paired comparison and... (part 10)

This is the right comparison for bounded, often non-Gaussian accuracy deltas, but it is not assumption-free. The test reports the primary matched-pairs rank-biserial effect $r_{\mathrm{rb}} = (T^{+} - T^{-})/(T^{+} + T^{-})$. We retain the monotone secondary transform $d_{\mathrm{eq}} = 2r_{\mathrm{rb}}/\sqrt{1 - r_{\mathrm{rb}}^{2}}$,

Paired comparison and... (part 11)

labeled a *rank-biserial-derived d-equivalent*, never raw Cohen's d and never a replacement for the mean contrast.

Paired comparison and... (part 12)

When r_{rb} saturates at ± 1, the transform diverges; tables print a signed saturation marker rather than a finite million-scale value. The primary-sweep headline display method RKL has rank-biserial effect 1.0000, d-equivalent saturated (r=+1) (large),

Paired comparison and... (part 13)

with the paired mean difference and bootstrap interval reported alongside it.

Bootstrap interval estimates

Every inferential mean — colony free energy, learning-curve KL, per-method accuracy, and the robust-minus-naive accuracy difference — carries a 95% percentile-bootstrap confidence interval (Efron and Tibshirani 1993) from `statistics.bootstrap_ci`, resampled from the declared unit. For the review grid,

Bootstrap interval estimates (part 2)

the unit is the seed after the nested trials have been reduced; for the primary fixed-world sweep, the interval is explicitly trial-level and conditional; for single-seed diagnostics, it is descriptive or trial-level.

Bootstrap interval estimates (part 3)

The single-colony mechanistic rate table is $n = 1$ per cell and is descriptive, not an inferential mean; its companion profile uses the declared nested design. This separation follows simulation-reporting guidance to declare the estimand and the independent Monte Carlo unit (Morris et al. 2019).

Bootstrap interval estimates (part 4)

The interval quantifies variation over the declared resampling unit (seed, recorded step, or matched trial) and is conditional on this simulation design, not on unmodeled real-world deployment uncertainty or alternative contamination models.

Bootstrap interval estimates (part 5)

The headline mean robust-minus-naive accuracy difference is 0.0809 with 95% CI $[0.0783, 0.0834]$.

At the verdict rate, the naive pool has mean accuracy 0.9021 (95% CI $[0.8993, 0.9049]$).

Bootstrap interval estimates (part 6)

The headline display method RKL has mean accuracy 0.9829 (95% CI $[0.9827, 0.9832]$).

Bootstrap interval estimates (part 7)

For every multi-seed summary we also report the Monte Carlo standard error ($\mathrm{MCSE} = s/\sqrt{n_{\mathrm{seed}}}$) and an approximate two-sided minimum detectable effect (MDE) at the configured target power. These are precision diagnostics over independent seeds,

Bootstrap interval estimates (part 8)

not claims about sampling a real population.

The MDE uses a normal approximation conditional on the observed seed-level standard deviation (Koehler et al. 2009); it is reported alongside the non-parametric interval rather than used to replace the paired test.

Multiple-testing deflation by BH-FDR

The sweep compares each robust divergence against the naive pool, so an uncorrected $p < 0.05$ across the family would manufacture false discoveries. We control the false-discovery rate with the Benjamini-Hochberg step-up procedure (Benjamini and Hochberg 1995) (`statistics.bh_fdr`, ISC-29) at $\alpha = 0.05$.

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Multiple-testing deflation by... (part 2)

The verdict family owns one robust-versus-naive contrast per robust method at the predeclared verdict rate; each rate table owns its own within-method rate family. Families are not pooled across figures or across the review-grid cells.

Multiple-testing deflation by... (part 3)

The review-grid families retain all configured method contrasts and do not derive a selected member by pooled mean.

The procedure returns both the rejection mask and the monotone BH q-values.

Multiple-testing deflation by... (part 4)

BH controls expected false discovery proportion within the stated family, not the family-wise probability of any false positive; the manuscript therefore states the family each table uses. The positive-contrast rule is strict and conjunctive:

Multiple-testing deflation by... (part 5)

a method is a BH-rejected positive contrast *iff* BH rejects its null **and** its rank-biserial effect is positive. This is a statistical decision for the named family, not a unique scientific winner.

Multiple-testing deflation by... (part 6)

The primary-sweep headline display method has raw $p = 1.11 \times 10^{-158}$, deflating to BH $q = 1.11 \times 10^{-158}$.

Prospective power analysis for the verdict rate

We report not only *that* the verdict rejected the null but the observed-effect planning power implied by the run and the number of pairs a confirmatory run could budget.

Prospective power analysis... (part 2)

The observed-effect design power of the headline paired Wilcoxon at the primary sweep's matched-trial unit, $n_{\text{trial}} = 960$, $\alpha = 0.05$, alternative `greater`, is 1.0000;

Prospective power analysis... (part 3)

the prospective sample size for the target power 0.80 at the observed effect is $n_{\text{trial}} = 5$.

Prospective power analysis... (part 4)

The estimator is the deterministic noncentral-normal approximation to the matched-pairs t-test, deflated by the Wilcoxon's Pitman asymptotic relative efficiency $3/\pi$,

Prospective power analysis... (part 5)

so the reported power is approximately calibrated for the signed-rank test the harness runs (the deflation is exact under a normal-shift alternative); note the power computation is directional for planning, while the reported p-values are two-sided.

Prospective power analysis... (part 6)

This is an observed-effect planning approximation conditional on the observed effect size, not independent evidence for the result and not a confirmatory power guarantee.

Reporting tables and the honesty boundary

The results sections render six statistics tables straight from these tokens: the per-rate accuracy sweep (tbl. 4), the robust-versus-naive verdict (tbl. 7),

Reporting tables and the... (part 2)

the standardized-effect estimates (tbl. 8), their paired inference and observed-effect planning diagnostics (tbl. 9),

Reporting tables and the... (part 3)

the per-method accuracy with bootstrap CIs at the verdict rate (tbl. 10),

Reporting tables and the... (part 4)

and the effect and inference projections of the per-contamination-rate paired tests (tbls. 5, 6).

Every cell is a generated token, never hand-typed.

Reporting tables and the... (part 5)

The honesty contract binds at exactly this tier. The effect-size, CI, and power enrichment above decorates the *server-side* `robust_aggregate` divergence-reweighting contrasts only.

Reporting tables and the... (part 6)

It does not certify the per-client β/rcce generalized-Bayes (FedGVI) guarantees, which are pinned by the locked core (Proposition 4 of sec. 9) rather than by these aggregation-level statistics —

Reporting tables and the... (part 7)

the three robustness axes of sec. 6.3 kept distinct.

Evidence classes and replication hierarchy

The reporting map in fig. 9 makes the reporting rule explicit. Read each numbered row across its six fields: result and class, estimand and unit, independent unit, nesting,

Evidence classes and... (part 2)

permitted support, and prohibited support. A formal identity, conditional server contrast, protocol check, and application receipt therefore cannot silently exchange evidentiary standing.

Evidence classes and... (part 3)

The top nesting strip supplies a shared reading grammar. A seed is independent only where its report declares it so; matched trials, agents, contamination roles, ordered steps, and posterior states otherwise remain nested. The three no-transfer boxes keep these ownership boundaries explicit.

Evidence classes and... (part 4)

The external source-dataset lane stays open until a frozen design and source-bound confirmatory report exists. Fourteen rows cover the headline result families and public software boundaries;

Evidence classes and… (part 5)

exact estimates and intervals remain in their typed reports and the native-unit cross-study summary (fig. 33).

Evidence classes and... (part 6)

FE: Formal / executable. proved or exactly tested in the project model.

Evidence classes and… (part 7)

SC: Source-conditional. inherits only under the cited paper's assumptions.

Evidence classes and... (part 8)

CE: Conditional empirical. estimated in the declared finite design.

Evidence classes and… (part 9)

SI: Scoped implementation. checks one software or protocol boundary.

Evidence classes and... (part 10)

O: Open. not established by the current evidence.

Evidence classes and... (part 11)

1. Project identity at c=0 [FE]
Estimand / unit: paired outputs
[categorical posterior]

Evidence classes and... (part 12)

1. Project identity at c=0 [FE]
Independent unit: exact identity.
Nesting: input → paired functions

Evidence classes and... (part 13)

1. Project identity at c=0 [FE]
May support: project limit. Does not support: full source protocol

Evidence classes and... (part 14)

2. Client-loss robustness [SC] Estimand / unit: generalized posterior [client update]

Evidence classes and… (part 15)

2. Client-loss robustness [SC] Independent unit: theorem; no empirical unit. Nesting: assumptions → update → posterior

Evidence classes and… (part 16)

2. Client-loss robustness [SC]
May support: source theorem. Does not support: server theorem; performance; calibration

Evidence classes and… (part 17)

3. Paired free-energy contrast [CE]
Estimand / unit: isolated − shared colony F [nats]

Evidence classes and… (part 18)

3. Paired free-energy contrast [CE]
Independent unit: configured seed.
Nesting: study → seed → condition → agents

Evidence classes and… (part 19)

3. Paired free-energy contrast [CE] May support: paired protocol contrast. Does not support: general benefit; exact replication

Evidence classes and... (part 20)

4. Dirichlet likelihood learning [CE]
Estimand / unit: KL(true A || learned A) [nats]

Evidence classes and… (part 21)

4. Dirichlet likelihood learning [CE]
Independent unit: configured seed.
Nesting: study → seed → steps → A rows

Evidence classes and... (part 22)

4. Dirichlet likelihood learning [CE]
May support: executed conjugate path. Does not support: universal rate; continuous model

Evidence classes and... (part 23)

5. Configured BMR sign control [FE]
Estimand / unit: pruning delta F [nats]

Evidence classes and… (part 24)

5. Configured BMR sign control [FE] Independent unit: closed-form comparison. Nesting: posterior → pruning → signed delta

Evidence classes and... (part 25)

5. Configured BMR sign control [FE] May support: configured sign control. Does not support: universal emergence

Evidence classes and... (part 26)

6. Exploratory client baseline [CE]
Estimand / unit: joint-config accuracy [fraction]

Evidence classes and... (part 27)

6. Exploratory client baseline [CE]
Independent unit: synthetic seed.
Nesting: sweep → seed → client → row

Evidence classes and... (part 28)

6. Exploratory client baseline [CE]
May support: composite configuration contrast. Does not support: RCCE-only; calibration; BNN claims

Evidence classes and… (part 29)

7. Server reweighting heuristic [CE] Estimand / unit: robust − naive [p or fraction]

Evidence classes and… (part 30)

7. Server reweighting heuristic [CE] Independent unit: seed or matched trial. Nesting: study → seed/world → trial → agents

Evidence classes and… (part 31)

7. Server reweighting heuristic [CE] May support: finite mechanism behavior. Does not support: influence bound; Byzantine tol.; universal win

Evidence classes and… (part 32)

8. Variational server rule [FE] Estimand / unit: descent, weights, transfer [mixed]

Evidence classes and… (part 33)

8. Variational server rule [FE] Independent unit: identity or seeded path. Nesting: config → start → iterates

Evidence classes and... (part 34)

8. Variational server rule [FE]
May support: objective + tested transfer. Does not support: global or universal claim

Evidence classes and... (part 35)

9. Moving / disjoint-FOV contrasts [CE]
Estimand / unit: policy/sharing [fraction, nats, steps]

Evidence classes and… (part 36)

9. Moving / disjoint-FOV contrasts [CE]
Independent unit: seed or fixed world. Nesting: world → seed → condition → agents/steps

Evidence classes and... (part 37)

9. Moving / disjoint-FOV contrasts [CE]
May support: executed-world behavior. Does not support: universal claim; physical multi-host

Evidence classes and... (part 38)

10. Hierarchy / sensitivity diagnostics [CE]
Estimand / unit: posterior/gap/gate [probability, fraction, nats]

Evidence classes and... (part 39)

10. Hierarchy / sensitivity diagnostics [CE]
Independent unit: seed or fixed diagnostic. Nesting: world → seed → level → agents/states

Evidence classes and... (part 40)

10. Hierarchy / sensitivity diagnostics [CE]
May support: executed hierarchy design. Does not support: universal benefit, depth, or emergence

Evidence classes and… (part 41)

11. Acuity recovery diagnostic [CE]
Estimand / unit: acuity, error, R^2
[p or unitless]

Evidence classes and… (part 42)

11. Acuity recovery diagnostic [CE] Independent unit: trial within acuity cell. Nesting: grid point → trial → rows → estimate

Evidence classes and... (part 43)

11. Acuity recovery diagnostic [CE]
May support: finite-grid recovery.
Does not support: global structural identifiability

Evidence classes and... (part 44)

12. Protocol-v1 / replay integrity
[SI]
Estimand / unit: frame/replay match
[frame or round]

Evidence classes and... (part 45)

12. Protocol-v1 / replay integrity
[SI]
Independent unit: local round.
Nesting: round → workers → frames → fields

Evidence classes and… (part 46)

12. Protocol-v1 / replay integrity [SI]
May support: protocol-v1 local integrity. Does not support: encryption/mTLS; multi-host; Byzantine tol.

Evidence classes and... (part 47)

13. Application receipt [SI] Estimand / unit: hash/source agreement [artifact]

Evidence classes and… (part 48)

13. Application receipt [SI] Independent unit: application invocation. Nesting: invocation → artifacts → levels

Evidence classes and… (part 49)

13. Application receipt [SI]
May support: hashes; requested source match. Does not support: validity; decision; acceptance

Evidence classes and… (part 50)

14. External calibration / confirmation [O]
Estimand / unit: paired log score [nats/observation]

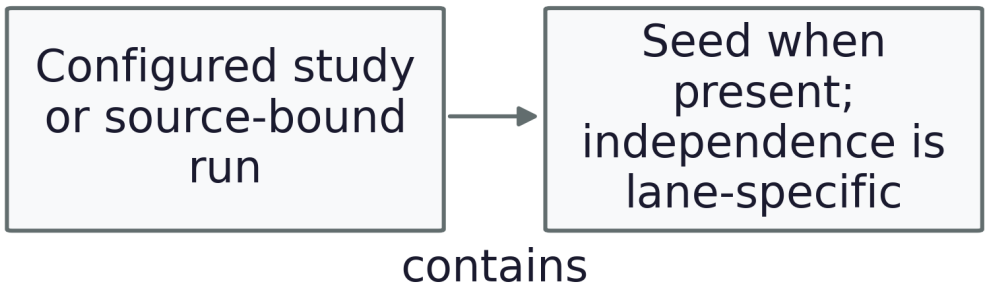
Evidence classes and… (part 51)

14. External calibration / confirmation [O]
Independent unit: dataset highest; seeds nested. Nesting: dataset → seed → clients → rows

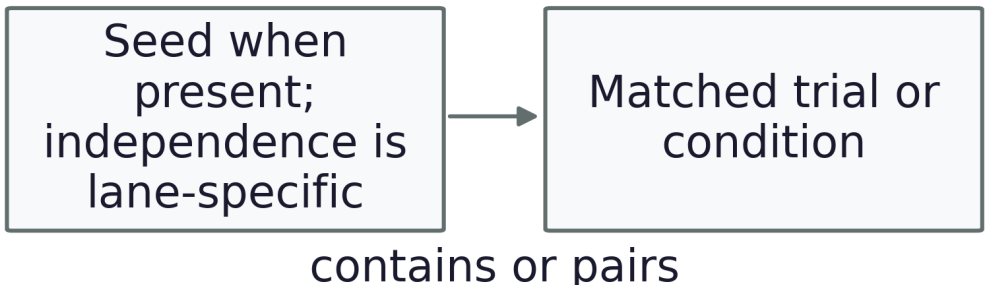
Evidence classes and… (part 52)

14. External calibration / confirmation [O]
May support: none before design + report. Does not support: retrospective or pooled-dataset claim

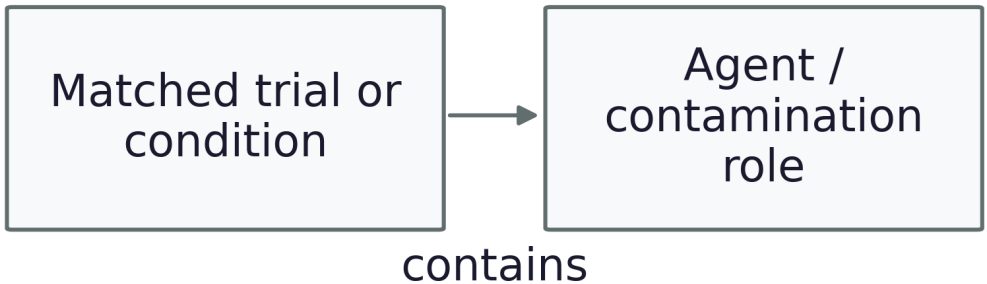
Evidence classes and... (part 53)



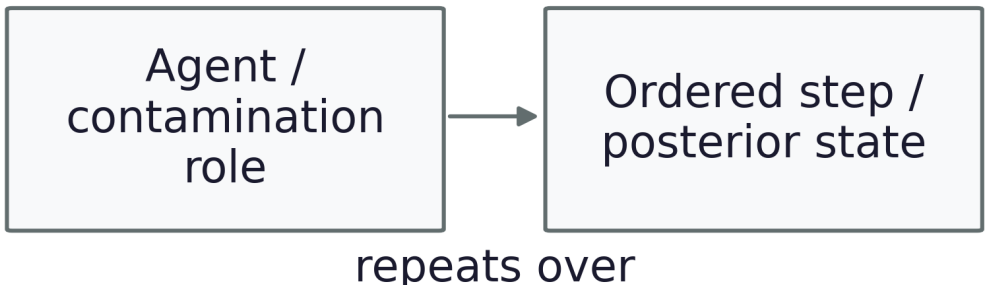
Evidence classes and… (part 54)



Evidence classes and… (part 55)



Evidence classes and… (part 56)



Evidence classes and… (part 57)

Nesting grammar: arrows mean contains, pairs, or repeats. Use only the levels declared in each evidence row.

Evidence classes and… (part 58)

NT1: guarantees do not transfer between client, server, protocol, and application lanes

Evidence classes and… (part 59)

NT2: nested agents, trials, and time steps are not promoted to independent datasets

Evidence classes and… (part 60)

NT3: open external confirmation is not implied by formal or software evidence

Computational complexity and scaling diagnostic

The release now reports computational complexity from the implementation itself. Let N be the number of agents, S the number of categorical states, I the solver-iteration budget, B the number of variational starts,

Computational complexity... (part 2)

and M the number of conditionally independent observation modalities.

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Computational complexity... (part 3)

The dominant dense work and the storage actually retained by the current NumPy paths are summarized separately so neither quantity is compressed at presentation scale. Here `LOO` means self-excluding leave-one-out sharing, and `server round` excludes queue and network wait.

Computational complexity... (part 4)

Time, aggregation. The log, robust, and variational pools have orders $\Theta(NS)$, $\Theta(INS)$, and $\Theta(BINS)$, respectively.

Computational complexity... (part 5)

Time, sharing and inference. Naive LOO, robust LOO, state inference, and a server round have orders $\Theta(N^2S)$, $\Theta(IN^2S)$, $\Theta(MS)$, and $\Theta(NlogN + INS)$, respectively.

Computational complexity... (part 6)

Storage, aggregation. The log, robust, and variational pools retain or peak at $\Theta(NS)$, $\Theta(NS + IS)$, and $\Theta(NS + IS + BS)$, respectively.

Computational complexity... (part 7)

Storage, sharing and inference. Naive LOO, robust LOO, state inference, and a server round retain or peak at $\Theta(NS)$, $\Theta(NS + IS)$, $\Theta(S)$, and $\Theta(NS)$, respectively.

Computational complexity... (part 8)

These are dominant interaction counts, not hardware-independent FLOP totals. The N^2 sharing term is material: with sensory attenuation enabled, one round computes one global pool and one leave-one-out pool per agent.

Computational complexity... (part 9)

The iterative server rules additionally retain their returned per-iteration histories, which is why their storage rows include IS. The server row includes worker-ID sorting and dense aggregation; incoming serialized belief volume is linear in NS,

Computational complexity... (part 10)

while queue and network latency are deliberately outside this local-compute accounting.

Computational complexity... (part 11)

The local path serializes one consensus-plus-weight result; if physical broadcast bytes are counted per recipient, that outgoing volume additionally has an N^2 term because each result carries the N agent weights.

Computational complexity... (part 12)

The accompanying seeded benchmark measures the real public call paths on the declared grids $N \in \{4, 8, 16, 32, 64\}$, $S \in \{256, 512, 1024, 2048, 4096\}$, self-excluding sharing $N \in \{4, 8, 16, 32\}$, and $M \in \{1, 2, 4, 8\}$.

Computational complexity... (part 13)

The fixed dimensions are $N = 256$, $S = 64$ for aggregation, and $S = 16384$ for state inference; direct aggregation and server timings use $I = 6$, while the public robust self-excluding sharing path uses its solver budget $I = 32$; the default variational path uses $B = 3$.

Computational complexity... (part 14)

Each grid point is warmed up 1 time(s) and measured 5 time(s), with median time plotted and the observed repeat range shown as a min–max bar.

The fixed input seed is 20260728 on the $arm64$ machine using Python $3.14.9$ and NumPy $2.4.2$.

Computational complexity... (part 15)

The measured log–log slopes are descriptive checks of the expected orders, not performance guarantees: agent-axis slopes are 0.94 (log-linear), 0.90 (iterative robust), 0.78 (variational), 1.64 (naive self-excluding sharing), and 1.91 (robust self-excluding sharing); state-axis slopes are 0.98, 0.96, and 0.96;

Computational complexity... (part 16)

the modality-axis inference slope is 0.72.

The slope fit is a timing diagnostic on this machine, not an inferential test and not evidence that the same constants hold under another BLAS, accelerator, process topology, or distributed network.

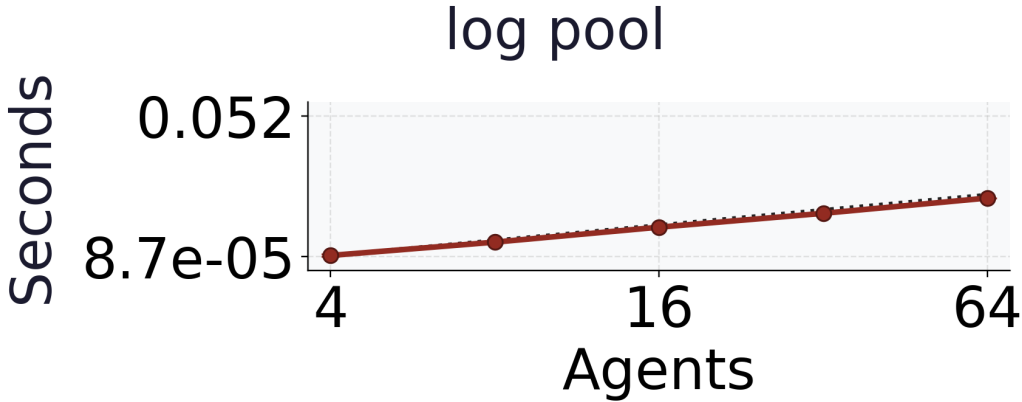
Computational complexity... (part 17)

A finite grid can also yield a sublinear fitted slope when validation, allocation, cache, and interpreter overheads are material; the implementation-derived order is the governing claim, not equality between a finite-grid slope and its exponent.

Computational complexity... (part 18)

The timing plot in fig. 10 visualizes the implementation-derived orders and the corresponding finite-grid timing diagnostic.

Computational complexity... (part 19)

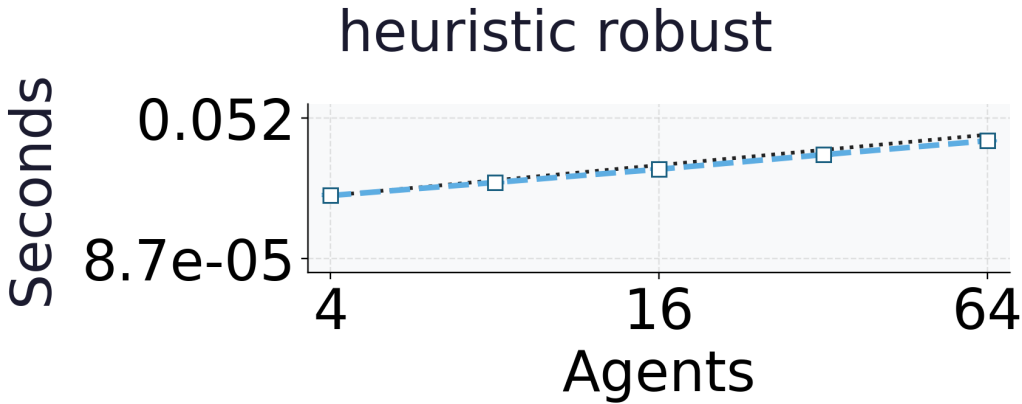


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Computational complexity... (part 20)

log pool by agents. Expected exponent: 1. Observed log-log slope: 0.944.

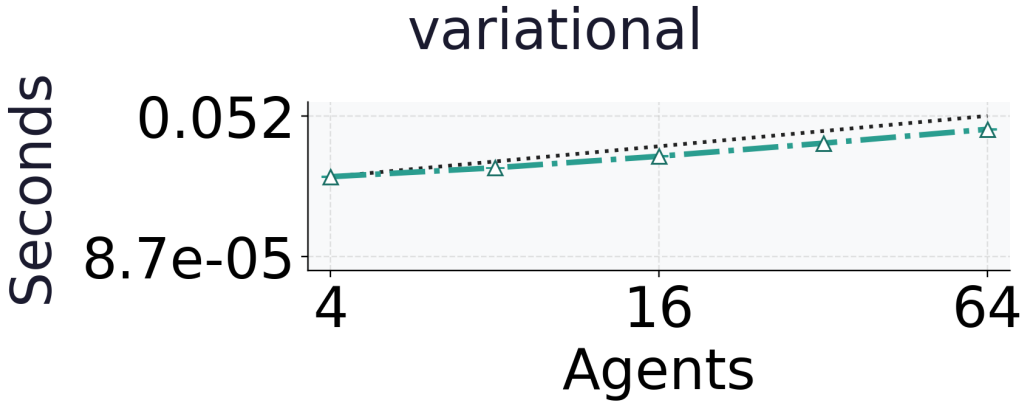
Computational complexity... (part 21)



Computational complexity... (part 22)

heuristic robust by agents. Expected exponent: 1. Observed log-log slope: 0.9027.

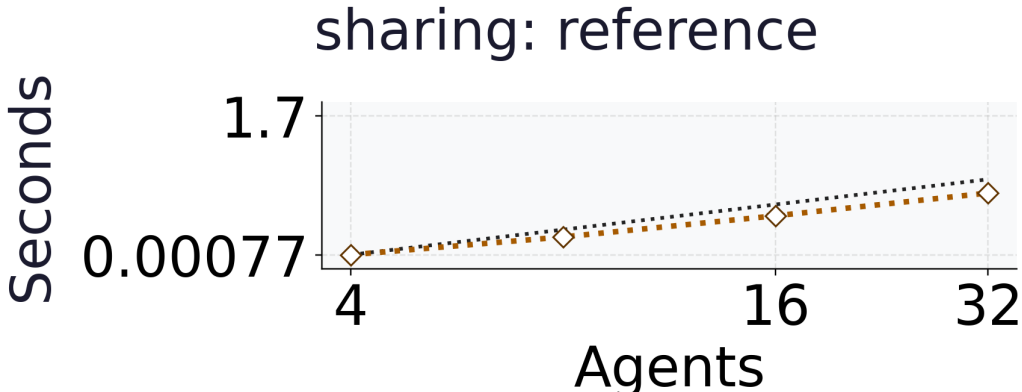
Computational complexity... (part 23)



Computational complexity... (part 24)

variational by agents. Expected exponent: 1. Observed log-log slope: 0.7848.

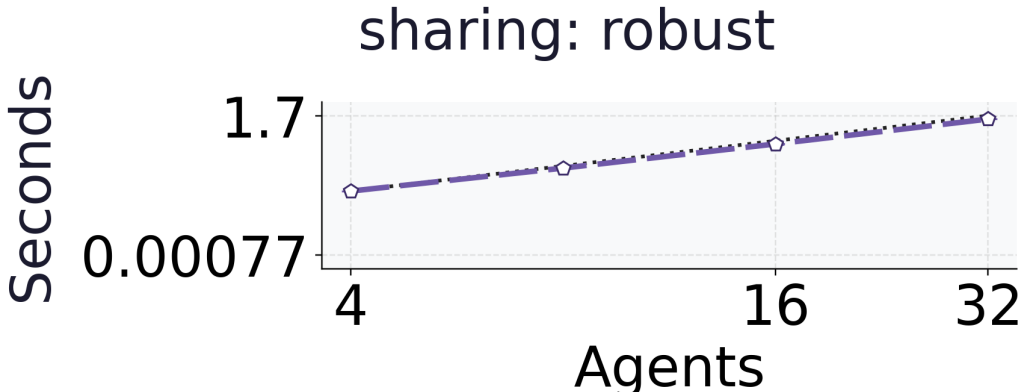
Computational complexity... (part 25)



Computational complexity... (part 26)

sharing: reference by agents. Expected exponent: 2. Observed log-log slope: 1.641.

Computational complexity... (part 27)

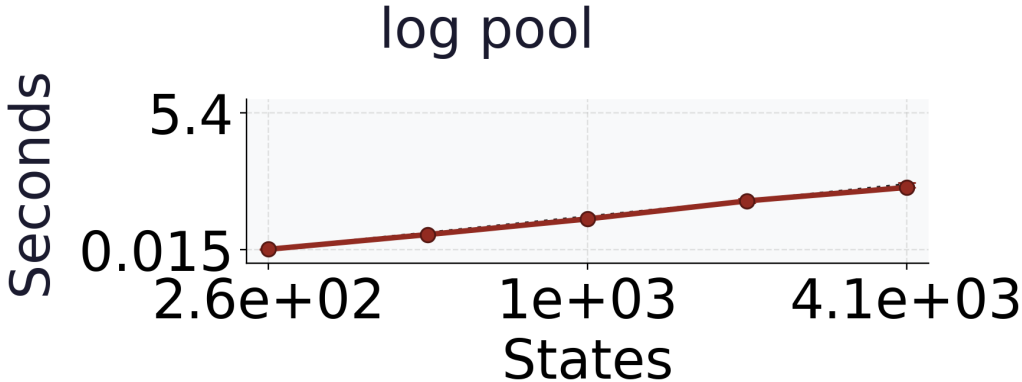


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Computational complexity... (part 28)

sharing: robust by agents. Expected exponent: 2. Observed log-log slope: 1.91.

Computational complexity... (part 29)

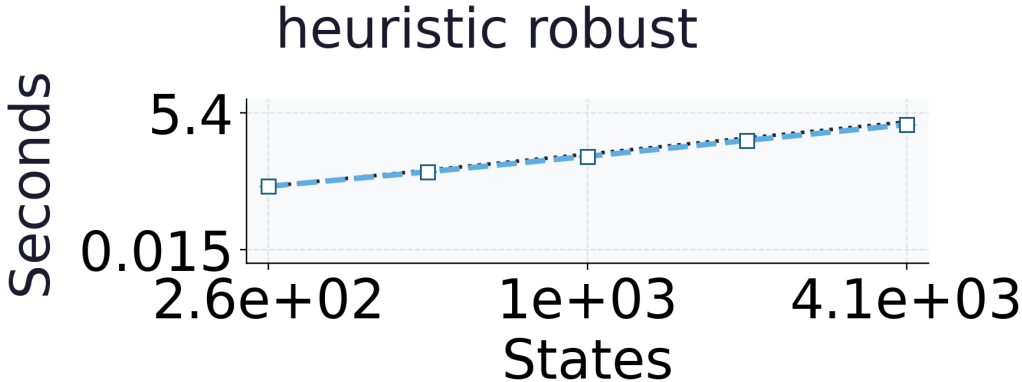


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Computational complexity... (part 30)

log pool by states. Expected exponent: 1. Observed log-log slope: 0.9762.

Computational complexity... (part 31)

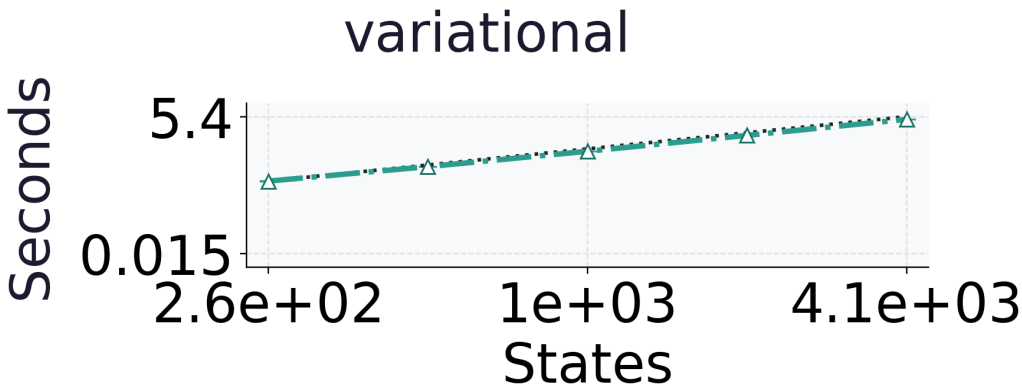


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Computational complexity... (part 32)

heuristic robust by states. Expected exponent: 1. Observed log-log slope: 0.9599.

Computational complexity... (part 33)

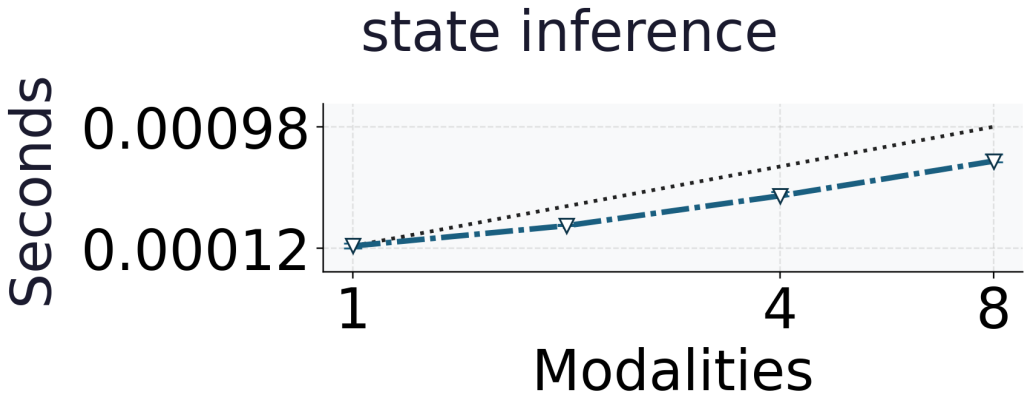


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Computational complexity... (part 34)

variational by states. Expected exponent: 1. Observed log-log slope: 0.9605.

Computational complexity... (part 35)



Computational complexity... (part 36)

state inference by modalities. Expected exponent: 1. Observed log-log slope: 0.7159.

Computational complexity… (part 37)

Medians over 5 timing repeats; min–max bars are observed ranges, not confidence intervals. Normalized dotted guides are symbolic-order references. Seed 20260728.

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